

The Coronavirus Disease of 2019

Machine Learning Project Report

CP468

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1. Abstract

The coronavirus disease 2019 (COVID-19) pandemic caused by the SARS-Cov2 virus has become the world's most serious and contentious health issue. It is associated with a variety of clinical manifestations as well as a high mortality rate. Estimating the number of total cases and identifying outcome predictors are crucial for countries to counter the suffering from high cases of COVID. To reduce the complexity of disease phenotypes, multivariate and machine learning methods can be used to develop prediction models. Machine learning algorithms have long been used in many applications that require the detection of adverse risk factors. Computational methods for machine learning (ML) have demonstrated their functionality in predicting potential outcomes to carry out informed decisions. The ability of ML modeling to predict the number of individuals affected by COVID-19 as a potential threat to humans is demonstrated in this study.

Keywords: Covid-19, Machine Learning Algorithms, Multivariate

2. List of Abbreviations

- ML - Machine Learning
- DL - Deep Learning
- MLRM - Multiple Linear Regression Model
- COVID -19 - Coronavirus Disease of 2019
- JHU - John Hopkin's University
- CSSE - The Center for Systems Science and Engineering

3. Introduction

In December 2019, Wuhan, China, reported an outbreak of a highly infectious respiratory disease that was later identified as Coronavirus Disease 2019. COVID-19 has since spread throughout the world, affecting many people's lives in various ways. To combat the COVID-19 pandemic, researchers created machine learning models to help predict outbreaks, allowing the medical system and government to prepare and allocate resources ahead of time.

In times of crisis, the effectiveness of policymakers' decision-making is heavily reliant on their ability to integrate and interpret the information. The COVID-19 crisis forces governments to make difficult decisions in the name of public health and safety. Essentially, policymakers must respond to an unknown threat while making decisions under time constraints and in the face of enormous uncertainty. The stakes are high, the situation is complex, and several interests, including (mental) health, the economy, and human rights, must be cautiously balanced.

The coronavirus affects everyone on the planet, either directly or indirectly. Someone can be directly confronted when the virus attacks them, and others may be indirectly affected due to the closure of their businesses, work, and daily activities. The spread of this disease has had significant impacts on the global economy. It affects all countries, whether they are developing, developed, or under development. Our goal is to assist in the prediction of the total number of future cases so that people can protect themselves and prepare accordingly. The model will also present the amount of effect each independent variable will have on the total number of cases which will allow the government to make better-informed decisions.

4. Materials & Methods

4.1. Dataset

This study's COVID-19 dataset is a collection of COVID-19 data maintained by Our World in Data. Throughout the COVID-19 pandemic, the data is updated daily and includes figures on vaccinations, tests, positivity, confirmed cases, confirmed deaths, GDP per capita, policy responses, and other variables of interest for a total of 218 countries.

The Center for Systems Science and Engineering (CSSE) at Johns Hopkins University created the dataset from the COVID-19 Data Repository (JHU). A team at Johns Hopkins University's Center for Systems Science and Engineering maintains the dashboard and dataset (CSSE). Since January 22, 2020, it has been publishing updates on confirmed cases and deaths for all countries. This has enabled millions of people all over the world to follow the pandemic's progress.

JHU's data is updated several times per day. This information is derived from governments, and national/subnational agencies all over the world; a complete list of data sources for each country is available on the Johns Hopkins GitHub site. It also makes its data public on the site.

4.2. Supervised Machine Learning Models

Supervised learning algorithms attempt to model the relationships and dependencies between the target prediction output and the input features in order to predict the output values for new data based on the relationships learned from previous data sets.

4.2.1. Multiple Linear Regression Model

Multiple Linear Regression is a machine learning algorithm that uses multiple independent variables to predict a single dependent variable.

The structure of a multiple linear regression model is as follows:

$$y = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \beta_0$$

Where...

Y : response variable

N : number of features

X_n : n-th feature

β_n : regression coefficient (weight) of the n-th feature

β_0 : y-intercept

$\beta_n x_n$: estimated slope-coefficient

4.3. Deep Learning Models

Deep learning is a machine learning technique that teaches computers to do what humans do instinctively: learn by doing. Deep learning is a key technology behind self-driving cars, allowing them to acknowledge a stop sign or distinguish between a pedestrian and a lamppost. It is essential for voice control in consumer devices such as phones, tablets, televisions, and hands-free speakers. Deep learning has received a lot of attention recently, and for good reason. It is achieving results that were previously unthinkable.

4.3.1. Recurrent Neural Network

5. Exploratory Data Analysis

5.1. Data cleansing

The process of detecting and correcting corrupt or inaccurate records from a record set, table, or database is known as data cleansing. It involves identifying incomplete, incorrect, inaccurate, or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.

Prior to implementing any models, the data was required to go through the process of data cleansing as there were multiple instances of NaN values, null values, and inconsistencies that would negatively affect the models. The data was first trimmed by

columns, then by rows. All columns that were numerical values were used as training data. The columns that were dropped include 'date', 'iso_code', 'continent', 'location', and 'tests_units'. These variables had NaN values and were not useful in the Machine Learning Model. The rows include data in the Country of Canada from 01/23/2020 - 06/26/2022. There were no outstanding outliers present in the dataset post-trimming.

5.2. Handling Null Values

Although missing values can be handled by deleting the rows or columns having null values, this was not ideal for the Machine Learning Model as it may decrease the accuracy of the model. The Null values in each column were replaced with the integer value 0.

6. Methodology

6.1. Introduction to Tools Utilized

Pandas - Python Library used to analyze data

Matplotlib.pyplot - a state-based matplotlib interface. It provides an implicit, MATLAB-like plotting method. It also serves as the figure GUI manager and opens figures on your screen.

NumPy - a Python library that adds support for large, multi-dimensional arrays and matrices, as well as a large collection of high-level mathematical functions to operate on these arrays.

Scikit-learn - a free machine learning software library for the Python programming language. It includes a variety of classification, regression, and clustering algorithms, as well as support vector machines.

6.2. Multiple Linear Regression Model

After the dataset had been trimmed, the remaining data consisted of 886 rows and 62 columns. The ultimate goal is to predict the number of future covid cases based off of the data present. In this analysis, the variable 'total_cases' is the dependent variable while the 62 features make up the independent variable.

6.2.1. Data Pre-Processing

Data preprocessing is the process of preparing raw data for use with a machine learning model. It is the first and most important step in developing a machine learning model.

The following commands were used for importing the libraries and necessary tools:

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import math, time
import numpy as np
from sklearn import preprocessing, svm, metrics
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

The following code imports the data set:

```
df = pd.read_csv('owid-covid-data.csv')
```

The following code is used to split the data into training and test set:

```
train_x, test_x, train_y, test_y = train_test_split(features, targets, test_size=0.2, random_state=43)
```

80% of the data is used as training data while the remaining 20% is used as the test set. The size of the training data is 710, the size of the testing data is 178, and the number of features is 62.

6.2.2. Fitting the MLR to the training set model

Since the dataset is well prepared for training, the regression model will need to fit the training set. The code will be as follows:

```
lr = LinearRegression()
lr.fit(train_x, train_y)
```

The model has been successfully trained using the training dataset. In the next step, the performance of the model will be tested using the test dataset.

6.2.3. Developing a Prediction of the test set

The final step for the model is checking its performance of the model. This will be done by predicting the test set result. For prediction, the `pred_y` vector is created which generates a large number of predictions for algorithm verification. Below is the code for it:

```
pred_y = lr.predict(test_x)
```

6.3. (Implementation of another Algorithm)

7. Algorithm Results

7.1. Results from the Multiple Linear Regress

7.1.1. Coefficients

The sign of a regression coefficient indicates whether the correlation between each independent variable and the dependent variable is positive or negative. A positive coefficient indicates that as the value of the independent variable increases, so does the mean of the dependent variable. A negative coefficient indicates that as the independent variable increases, the dependent variable decreases. The further the variable is from 0, the larger effect that variable has on the dependent variable; total_cases.

There are 62 features in our model and one coefficient for each variable. The new_people_vaccinated_smoothed_per_hundred has a -2516871.112 as its coefficient. This means that as the new confirmed cases of COVID-19 (7-day smoothed) per 1,000,000 people increase, the number of total cases decreases. The coefficient for the total_deaths_per_million is 2084763.449. This is the highest positive variable that causes the total_cases to increase. This may be due to the increase in reporting cases as deaths occur. This analysis allows government officials and policymakers to better analyze which variables and factors have a larger effect on the total_cases and what can be done to keep the number of total cases low.

The following are the coefficients for each independent variable from smallest coefficient value to the greatest:

41	new_people_vaccinated_smoothed_per_hundred	-2516871.112
8	new_cases_smoothed_per_million	-1531184.981
16	hosp_patients_per_million	-548573.616
11	new_deaths_smoothed_per_million	-365578.887
14	icu_patients_per_million	-141536.265
3	total_deaths	-54764.068
37	people_fully_vaccinated_per_hundred	-53607.203
27	positive_rate	-24961.762
38	total_boosters_per_hundred	-22548.508
1	new_cases	-20144.917
4	new_deaths	-7419.666
39	new_vaccinations_smoothed_per_million	-60.204
25	new_tests_smoothed	-33.822
28	tests_per_case	-33.715

42	stringency_index	-33.688
0	index	-31.919
21	total_tests	-7.009
29	total_vaccinations	-0.557
22	new_tests	-0.376
30	people_vaccinated	-0.064
58	excess_mortality_cumulative_absolute	-0.036
33	new_vaccinations	-0.006
20	weekly_hosp_admissions_per_million	-0.005
56	life_expectancy	-0.001
50	cardiovasc_death_rate	-0.001
45	median_age	-0.001
53	male_smokers	-0.001
43	population	-0.000
51	diabetes_prevalence	-0.000
47	aged_70_older	-0.000
52	female_smokers	-0.000
44	population_density	-0.000
49	extreme_poverty	-0.000
54	handwashing_facilities	0.000
57	human_development_index	0.000
55	hospital_beds_per_thousand	0.000
46	aged_65_older	0.000
19	weekly_hosp_admissions	0.011
18	weekly_icu_admissions_per_million	0.021
32	total_boosters	0.061
17	weekly_icu_admissions	0.089
31	people_fully_vaccinated	0.141
48	gdp_per_capita	0.481
34	new_vaccinations_smoothed	1.595
61	excess_mortality_cumulative_per_million	2.173
40	new_people_vaccinated_smoothed	6.596
59	excess_mortality_cumulative	16.392
60	excess_mortality	19.601
6	total_cases_per_million	38.051
12	reproduction_rate	3096.940
13	icu_patients	3725.303
5	new_deaths_smoothed	9688.578
24	new_tests_per_thousand	13930.890
15	hosp_patients	14405.930

```

36         people_vaccinated_per_hundred  24486.056
2         new_cases_smoothed  40225.230
35         total_vaccinations_per_hundred  211903.849
23         total_tests_per_thousand  266837.290
10         new_deaths_per_million  280315.782
7         new_cases_per_million  766879.881
26         new_tests_smoothed_per_thousand  1291330.774
9         total_deaths_per_million  2084763.449

```

7.1.2. Analysis of the Regression

```

print('Analysis of regression')
print('-----')
print(f'Score: {lr.score(features, targets)}')
print(f'R squared: {metrics.r2_score(test_y, pred_y)}')
print(f'Mean error: {metrics.mean_absolute_error(test_y, pred_y)} cases')
print('-----')

```

```

Analysis of regression
-----
Score: 0.9999746757931892
R squared: 0.9999726896454993
Mean error: 3057.015806166927 cases
-----

```

The r-squared value for this model is 0.99997 which means that the regression function fits the data well. The predictions made by the model are 99% true to the actual values according to the score of the Multiple Linear Regression Model above. The mean error is approximately 3057 cases which suggest an average of 3057 error cases for every prediction made.

7.1.3. Predicted Values

The following code provides the values predicted by the Machine Learning model for the following three days after 2022-06-28 :

```

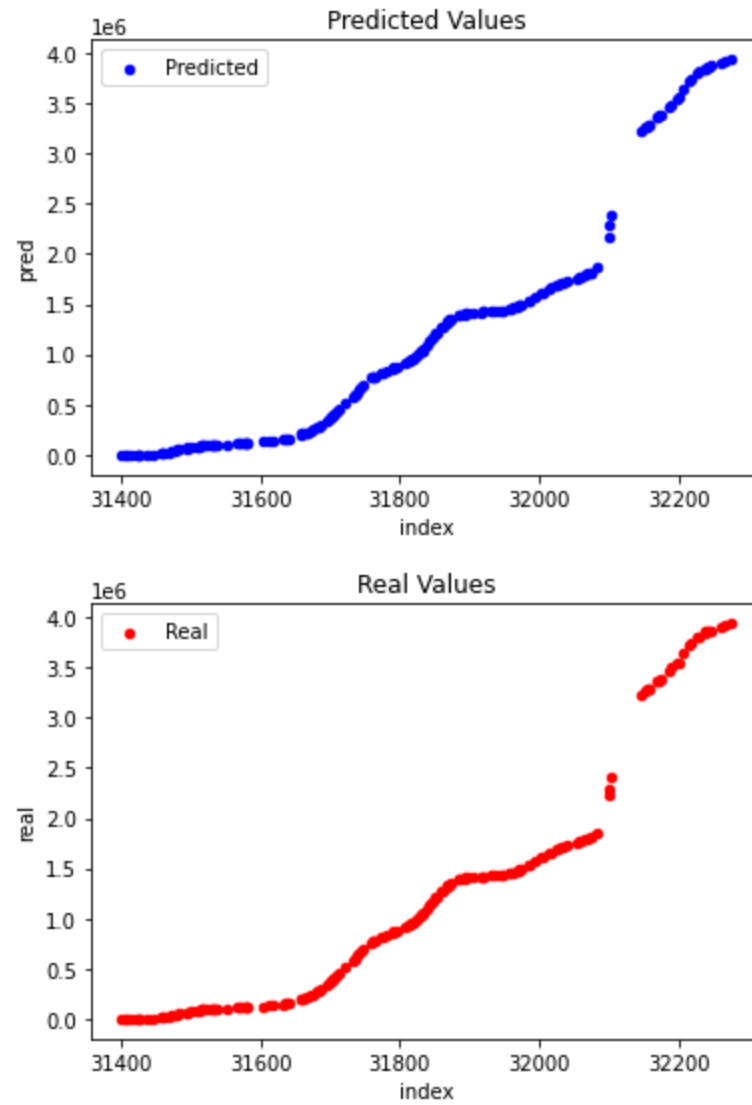
[7]: future_y = lr.predict(future_x)

for i in range(days_offset):
    print(f'Predicted cases {i+1} days after {dates.tolist()[-1]}: {int(future_y[-i])}')

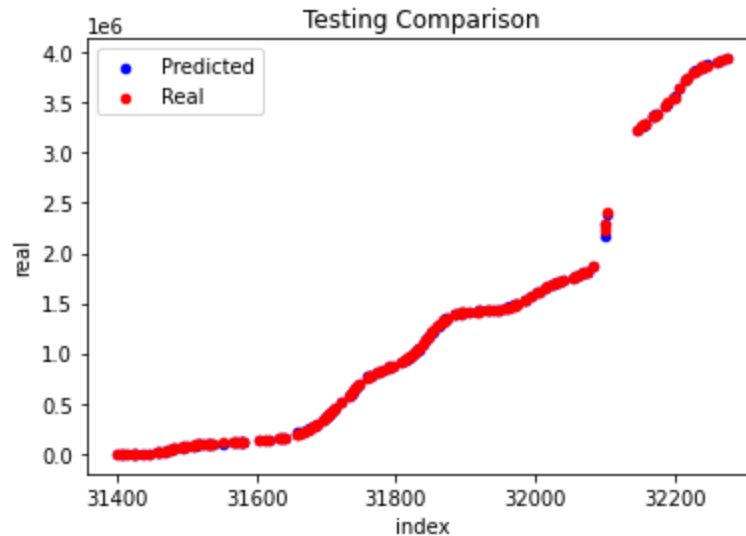
Predicted cases 1 days after 2022-06-28: 3931640
Predicted cases 2 days after 2022-06-28: 3935835
Predicted cases 3 days after 2022-06-28: 3932369

```

Below provides the graph of the predicted values from the ML model and the actual values from the data:



A testing comparison graph can be created with the two above graphs:



8. Conclusions

In Conclusion, the Multiple Linear Regression model provides an almost accurate graph in regards to the number of total cases on a given day. This information would be valuable for policymakers who make severely important decisions on a day to day basis as well as companies that may be affected by COVID. It would allow companies to prepare for surges in the virus and act accordingly.

8.1. Future Scope

In the future, this analysis may be improved by implementing more than one Machine Learning Model such as Recurrent Neural Networks (RNN). This would have allowed for more detailed analysis. The current model may be improved by replacing NULL values with an average value. The Multiple Linear Regression Model could also be tested with other countries as well. The results regarding the predictions and actual values would help in proving the accuracy of the Multiple Linear Regression Model.

9. References

Data Science - regression table: R-squared. Data Science Linear Regression R-Squared. (n.d.).

Retrieved July 18, 2022, from

https://www.w3schools.com/datascience/ds_linear_regression_rsquared.asp

Owid. (n.d.). *Covid-19-data/public/data at master · owid/covid-19-DATA*. GitHub. Retrieved

July 18, 2022, from <https://github.com/owid/covid-19-data/tree/master/public/data>

Ritchie, H., Mathieu, E., Rodés-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J.,

Macdonald, B., Beltekian, D., & Roser, M. (2020, March 5). *United States: Coronavirus*

pandemic country profile. Our World in Data. Retrieved July 18, 2022, from

<https://ourworldindata.org/coronavirus/country/united-states>